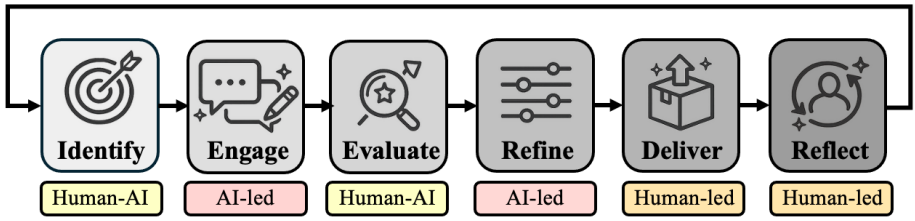
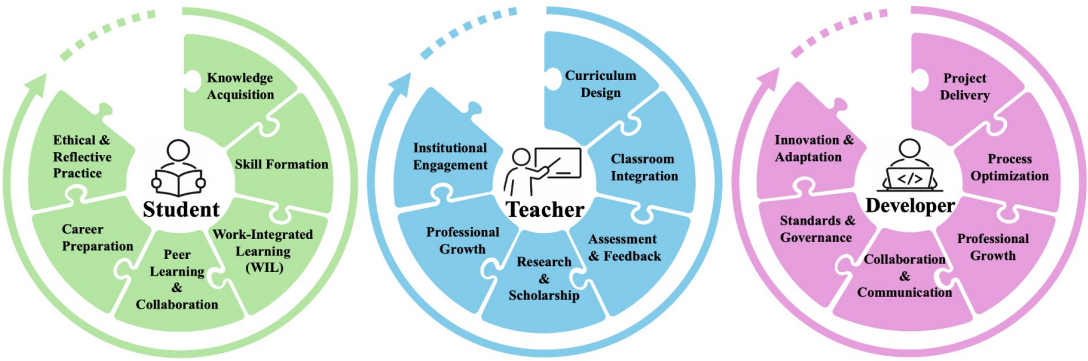
Layer 1. AI-integrated Task Pipeline

Pipeline



Actors	Identify	Engage	Evaluate	Refine	Deliver	Reflect
Student	Clarify learning goals	Perform learning tasks	Assess learning outcomes	Improve learning outputs	Complete learning tasks	Reflect on learning activities
Teacher	Define teaching goals	Design learning activities	Assess teaching quality	Improve teaching design	Deliver instructional materials	Reflect on teaching strategies
Developer	Define task requirements	Perform development tasks	Validate task outputs	Optimize task solutions	Release task deliverables	Reflect on task performances

Layer 2. Professional Development Loop



Layer 3. Computing Education Ecosystem

